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PMC-Sierra – B

A Case Study of Organizational Dynamics in a Rapidly Growing Firm

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As PMC-Sierra entered 1998, the business environment in which it operated became even more turbulent. Like the gold rush days in California and the Klondike over a century before, thousands of new entrants rushed to develop and use new networking technologies. Clearly, there were substantial advantages to staking claims for the best prospects early in the game, but no one knew exactly which technology, or combinations of methods, would prove to be durable features of this rapidly growing industry. Moreover, competition was both competitive and collaborative. While individual firms competed to see who could first develop a particular feature of new technology, these same firms, or groups or consortia of firms, collaborated in agreeing that certain industry standards would characterize their networking capabilities.

Current demand for established products and procedures expanded exponentially – as Internet volume doubled about every three months. Accurate forecasts for future demand were impossible to establish in such an uncertain environment. Nevertheless, there was a general consensus that firms who were not striving to remain at the forefront of this unprecedented expansionary opportunity had no chance to participate in and benefit from the huge payoffs anticipated from this industry.

The massive investments being made in network infrastructures meant huge potential for PMC-Sierra as their products were at the heart of the functioning of these networks. It also meant increased threats of competition from new upstarts and from established players. As PMC's senior managers contemplated their organization and its turbulent environment, they faced many difficult questions of how to manage their firm in light of current and emerging circumstances.

Market for Networking Products

One way to appreciate the status of the development of the Internet during 1998-2000 is consider the analogy of building a national or international highway system from the beginning. First, the actual technology of building roads is still in its infancy. The capacity of any highway to carry traffic at a particular speed and volume is known only in a limited way. Second, our understanding of how to build and develop highways is changing rapidly. Things we couldn't do a year ago are now quite feasible. However, some capacity problems remain relatively the same as before. Third, both the capacity and the potential speed of traffic are increasing rapidly and unevenly. Some innovations mean that on certain segments of the highway, traffic is slowed while on others it can move very rapidly. Fourth, the connections between various segments are often unbalanced. There's congestion at various points where traffic either enters or exits the main highway, and the capacity of both entry and exit routes also change rapidly. Fifth, overall patterns of future demand are largely unknown as increases in highway capacity influence the nature of future demand. Increased use may follow the provision of increased capacity – but there are also large, new sources for demand that may or may not emerge while the highway is being designed and built. Finally, there are many contractors attempting to build alternative highways and make use of existing networks. And so on.

The growth and development of the networking industry in 1998-1999 also possessed many of these characteristics. Ignoring the international features of networking, and focusing solely on the United States, some appreciation of the diversity, size, growth, and nature of interconnected product markets in the networking industry can be seen in the data presented in Table 1 below. The size of the overall market was in excess of US$120 Billion in 1999, and exhibiting an annual growth rate of just under 20 percent each year. Second, some market segments were increasing at astonishing rates, whereas others were declining. However, there was little stability to these patterns as technologies that grew in one year would not necessarily be in demand in the following year.

More detailed examination of particular product groups pertinent to PMC's perspective on this massive and dynamic market illustrates both the uncertainty and importance of strategic choice for firms attempting to prosper in this environment. The main customers for networking products were divisible on the basis of the type of network they were involved with. Broadly speaking, customers were either local area network (LAN) customers or they were wide area network (WAN) customers. By 2000, virtually every company, regardless of their size, had local

Table 1. U.S. Data Communications Market Profile

Revenues (Millions $)				% Change	
PRODUCTS	**1997**	**1998**	**1999**	**1997-1998**	**1998-1999**
Hubs	1789	941	826	(47)	(12)
Routers	3184	3900	4182	22	7
Workgroup and Lan Switches	2750	3081	3361	12	9
Campus Backbone Switches	1547	2310	3392	49	47
NICs	2115	2569	2684	21	4
Remote Access Servers	1604	1732	2104	8	21
Enterprise Frame Relay Switches	179	216	238	21	10
Enterprise ATM Switches	394	524	653	33	25
Frame Relay Access Devices	380	496	527	31	6
Modems	3077	3289	2796	7	(15)
Cable Service Equipment	66	125	248	89	98
DSL Equipment	33	77	139	133	81
Time-Division Multiplexers	590	573	568	(3)	(1)
Sonet/DWDM Equipment	3657	4335	4867	19	12
Core Carrier Data Switches	2570	3275	4136	27	26
PBXs	4805	6107	7277	27	19
Other	30901	35379	40906	14	16
PRODUCTS, TTL	**59641**	**68929**	**78904**	**16**	**14**
SERVICES	**27073**	**34355**	**44072**	**27**	**28**
PRODUCTS AND SERVICES TOTAL	**86714**	**103284**	**122976**	**19**	**19**

area networks (LANs) that linked together their computers and other devices, including printers. Just as the Internet had risen in exponential popularity over the latter half of the 1990's, so did the demand for LAN technology. As companies began to realize the benefits of a shared LAN, users began to transmit greater amounts of information, leading these organizations to clamor for higher bandwidth solutions.

At first the demand came from data intensive corporations, including banks, stock exchanges, and the government. As the Internet's popularity rose, though, mid-size organizations began to invest in their own networks, and upgraded their network backbones to speeds greater than 10 Megabits per seconds (Mbit/s). Two competing technologies vied for this market: Fast Ethernet (defined as 100 Mbit/s) and ATM technology (155 Mbit/s). There was also a 622 Mbit/s ATM technology that was reserved for companies with extremely high traffic volumes. Ethernet technology eventually surpassed the speeds possible with the 622 ATM with the introduction of Gigabit Ethernet (defined as 1000 Mbit/s).

The Wide Area Network (WAN), on the other hand, had a completely different set of customers. This group of buyers was responsible for managing communications between organizations and individual users over larger distances. Whereas a LAN was mainly housed within a single department or building, a WAN might cover a city, an entire country, or even reach across the world. Traditional telecom companies such as AT&T, Sprint, and Deutsche Telekom dominated this market, as did newer data networking upstarts such as UUNet, Williams, and Global Crossing. These companies relied on standardization, compatibility, and interconnectivity as their customers depended upon them to provide a dial tone and a connection to anywhere in the world, be it for a voice conversation, a facsimile transmission, or a video conference. Thus, these companies limited their purchases of necessary products to those that adhered to strict industry and technology standards.

The connection between a LAN and a WAN is handled by special computers known as routers. Routers are attached to a LAN – their purpose is to direct traffic between the local network and the rest of the Internet. An email, needing to leave the LAN, gets sent to a router, which then interprets the target's "address" and distributes the message accordingly. Likewise, when a message is destined for an individual user within the LAN, that same router or switch handles the incoming transmission and distributes it accordingly.

It was generally thought that the three key demand drivers during this time were the emergence of the Internet, the upgrade of corporate data networks, and remote access. The common element to each of these drivers was the high bandwidth demand for data. Rapid growth in the number of Internet users caused data traffic on public networks to explode. Data traffic was expected to triple each year until 2005, at which time the transmission of data was expected to overtake the transmission of voice as the dominant traffic type over the world's telecommunication networks.

As telecommunications networks shifted away from voice and towards data traffic, the technology for telecommunications equipment was also expected to change: away from circuit-switch technology to packet-based networks optimized for carrying data. Consequently, data networking equipment vendors such as PMC-Sierra were poised to benefit from participation in the burgeoning market for worldwide telecommunications equipment, which was expected to grow to $340 billion by 2002. Three areas were key to next-generation networks that were being built at the time: converged networks, wireless, and broadband.

1. Convergence refers to the transmission of voice, data, and video traffic over a single network. Convergence was still in early stages of development, but represented the fastest-growing market for telecom spending. In 1999, the market for hardware and software for computer/telephone integration surged by 66.2 percent. The acceptance of this technology was taking hold at a much faster rate than initially expected, injecting new growth into the networking and telecommunications equipment industry.

2. The growth of the wireless market was another critical driver of networking infrastructure building. Voice traffic initially fuelled the wireless build out. Approximately 283 million mobile phones were sold worldwide in 1999, up 65 percent over the previous year. In 1999, data accounted for only 5 percent of wireless traffic, and data services amounted to roughly $500 million. However, it was predicted that the wireless data services market could reach $60 to $80 billion worldwide by 2005.

3. The push for broadband Internet connections also added to the demand for increased investment in networking equipment. Broadband connections that were employed at the time included cable modems, DSL, satellite, and fiber optics. Jupiter Communications estimated that by 2002, 12 percent of North American households would have cable modem access, 6 percent would have DSL, and less than 2 percent would be using satellite.

The networking industry was not at all homogeneous. Particular firms focused their efforts on particular market segments they believed held most potential and for which their firm had a comparative advantage. A firm's focus on a particular market segment could be wildly successful or a dismal failure. Strategic choice of products and product markets was a critical feature of this industry. For example, through most of the 1990's the LAN and WAN segments of the networking industry were dominated by four major data networking companies: Cisco Systems Inc., 3Com Corp., Nortel Networks Corp., and Cabletron Systems Inc. By 1999, however, each of these companies emerged from the changes the Internet had brought to the networking industry with quite different strategies on how best to position themselves for the years and changes to come.

Nortel and Cisco had begun focusing on technologies designed for the network "backbone", a term used to describe the Internet's main network connections. The backbone carried the bulk of the Internet's traffic at incredibly high speeds, and was formed by the largest networking companies and Internet Service Providers (ISPs) such as MCI WorldCom, Sprint, GTE, and America Online. By connecting to each other at certain points, these networks formed a high-speed, high-bandwidth pipeline that crossed North America and extended to Europe, Japan, mainland Asia, and the rest of the world. On the other hand, 3Com began to emphasize networking equipment designed for the "edge" of the network, such as client productivity products. A further strategy was pursued by Cabletron Systems that reorganized itself into four different operating companies, each dedicated to markets inspired by opportunities presented by the Internet: service providers, professional services, infrastructure management, and enterprise e-business.

As the networking industry continued to expand, a huge variety of technical choices were being considered, attempted, and evaluated. For example, one way for the Internet to provide increased capacity was to provide increased bandwidth. This feature of the Internet allowed the simultaneous transmission of 'rich' data such as that associated with video. However, traditional networks designed for more 'lean' voice transmission required the network to break down and re-start Internet sessions several hundred times per hour. Older telecom networks were designed to set-up and tear down voice sessions lasting 10 minutes or longer and, as such, were inadequate to handle multiple Internet sessions as short as a `point-and-click'. The best ways to add bandwidth for the network included increasing the density of the bandwidth pipes and utilizing traffic management and congestion control techniques to make more effective use of the existing bandwidth pipes. Both trends were occurring simultaneously as the global networking infrastructure moved towards an equipment overhaul. However, building the physical infrastructure took time, and could only be used when there was a complete connection between fully serviced nodes of the network. In 1998 there was more than 100,000 miles of unused fiber bandwidth pipes deployed by carriers such as Williams, Level 3, UUNET, IXC, and 360 Networks.

These bandwidth suppliers used a variety of technologies to deliver their network services. Many of the traditional phone companies preferred "multi-service" Frame Relay and Asynchronous Transfer Mode (ATM) networks in order to leverage their existing infrastructures while providing the Quality of Service (QoS) required to support data, voice and other communication traffic. The young upstart networking companies were more data-centric, and preferred more Internet Protocol (IP) packet based data networks. The newest network deployments focused on transporting data, voice and other communication signals over a single multi-service network. Many carriers were building networks that converged (IP) layers over ATM and Frame Relay sub-networks or directly onto fiber.

Carriers could commit to particular service levels because the QoS traffic management techniques made possible by protocols such as ATM had developed an excellent capability for providing QoS standard guarantees. Protocols such as IP had less developed, but rapidly improving, QoS characteristics. The lack of acceptable IP-related QoS standards at that time prohibited the implementation of IP across the emerging global network. Thus, multi-service networks were still preferred by most of the traditional phone companies.

Nevertheless, many carriers believed the next generation of communications networks would be IP-based, as their networks would transport far more data than any other type of traffic. Eventually, carriers intended to incorporate the exemplary redundancy and reliability aspects of the voice network infrastructure in new next-generation packet networks because they recognized that voice and other traffic would still need to be transported. Consumers expected their telephones to work from the moment they picked up the receiver and heard the "dial tone" to the moment they hung it up, and they did not expect to "re-boot" their phones. In keeping with this performance expectation, the builders of the Internet were attempting to develop IP networks that offered the same levels of dependability.

Still more technological change was evidenced by the emergence of the Optical Transport Network (OTN) - a network based on glass and light transmission, rather than electrical transmission. This technical innovation was hailed as one of the greatest evolutions in the broadband-networking arena. The majority of the early Internet infrastructure was based on electrical transmission over copper wire. However, many carriers began to consider the massive bandwidth capacity provided by fiber optics and Dense Wave Division Multiplexing (DWDM), a method of using color wavelengths to create a new network "line" for every color, a more attractive investment for long distance transmissions. As such, the major networking equipment makers (Cisco, Nortel, and Lucent) began to invest heavily in this emerging technology. In fact, in a period spanning 9 months, the big three shelled out some $30 billion to buy startup companies whose products bridged the differences between older electronics technology and that of the emerging optical networks.

The emergence of the OTN resulted in an optical/electrical convergence where optical functions had to be meshed with electrically managed functions. In the emerging optical networks, optical wavelengths transported signals and electrical semiconductors managed the higher layer protocols, such as IP routing or switching. PMC products merged existing electrical-based communications protocols, such as ATM, SONET/SDH, Internet Protocol (IP) and Gigabit

Ethernet with new optical-based protocols. An area of research and development that was bearing fruit by 2000 was all-optical switching, a technology that would eliminate the need to transform optic signals to electrical signals in order to move data around the world's various networks. While many companies, like Lucent, pursued the use of mirrors for optical switches, Hewlett-Packard spun off Agilent, which used bubble jet printer technology to develop a completely different all optical switch.

The Semiconductor Industry and Stock Markets

The semiconductor business is notoriously cyclical, and earnings outlooks for chipmakers can change suddenly as new production capacity comes on stream. On stock markets, the severity of those shifts is magnified by huge flows of investors' money in and out of semiconductor stocks. Until the early March technology sell-off, shares of chip producers – and companies that make equipment used to manufacture chips – enjoyed an historic ascent. June 30, 2000 marked the 18th month of a torrid climb in semiconductor stock prices. The Philadelphia semiconductor index, the sector's key stock indicator, gained 602 percent between October 1998, and March 2000. Over that same time period, the broader, high-tech Nasdaq composite index climbed 256 percent.

The euphoria manifest in these astonishing evaluations of profit expectations in the networking industry followed a severe downturn in the industry between 1996 and 1998 when the Asian economic crisis, combined with a glut of PC chips, to dampen expenditures on new chip developments. A general turnaround in the world economy following that time led to the demand for chips far outstripping the capacity of chipmakers to supply their silicon wares. As a result, prices begin to rise as suppliers could afford to charge higher prices for scarce silicon. The capacity shortage prompted giants such as Intel Corp. and Texas Instruments Inc. to declare plans for new factories. The prospects for increased supply in turn led some analysts to downgrade their earnings forecasts for the semiconductor industry.

PMC was one of the big winners from the tech bubble of 1999-2000 and was one of the few companies that did not see its sky high valuation come immediately crashing to the ground. PMC's stock price, which had remained relatively flat until 1998, began to rise along with the market and peaked in March, 2000 at $250 a share after it had split two-for-one two months earlier. Even after the market crashed in the Fall of 2000, PMC shares were still trading in the $180 category, approximately a 2,000 percent gain from 1998. With these market evaluations, PMC-Sierra had become the highest valued company in British Columbia and one of the highest valued companies in Canada.

One consequence of these extraordinarily high stock evaluations was the implicit decline in PMC's cost of capital when it used its stock to purchase acquisitions. Acquisitions were a critical feature of PMC's overall corporate strategy as they permitted PMC to 'buy' the already developed technical skills available in acquired companies. Such purchases allowed PMC to grow and develop faster than if they relied on the recruitment and development of skilled staff alone. In addition, PMC was able to spread its portfolio of risk in this highly uncertain environment by targeting acquisitions that enabled PMC to occupy technical and market niches they had not been able to develop on their own.

Human Resource Issues

By the end of the 1990's, high tech companies struggled to find and attract star talent to their companies that were necessary to succeed in the Information Age. The value of corporations in the "New Economy" was no longer measured in terms of physical assets; rather, the value of high tech companies was determined by the quality of personnel an organization employed.

Talented labor was scarce and, as a number of research firms reported, an increasing number of high tech positions were going unfilled. The Information Technology Association of Canada (ITAC) estimated there were as many as 32,000 information and commerce jobs begging in Canada by the beginning of 2000, with vacancies expected to rise to 50,000 in the early years of the new millennium. A study undertaken by IDC (Canada) Ltd. revealed that some 60 percent of large Canadian companies had suffered project delays due to a shortage of IT professionals. The high tech worker shortage was not limited to North America either. According to the Frankfurt-based European Info Technology Observatory, there were some 600,000 IT jobs unfilled in the world in early 2000.

Though PMC's growth, rise in stock price, and desirable location were advantageous, PMC still faced a number of difficulties in their recruiting efforts. Canada's personal income-tax system was a source of considerable debate during the 1990's. Corporations and some politicians argued that high personal tax rates vis-à-vis the United States resulted in a "brain drain" of top talent from Canada to the U.S. The British Columbia maximum marginal tax rate of 54% percent (which the majority of high tech employees easily met, often soon after graduation from university) kicked in at roughly a taxable income of $63,000, comparing unfavorably to the top marginal rate of 43 percent on incomes in excess of $379,000 in the U.S. In addition to this, capital gains obtained from the sale of stock options were taxed at a rate of about 37 percent in Canada, versus 32 percent in the U.S. As a large portion of a high tech employee's yearly income was potentially derivable from the exercising of stock options, a high capital gains taxation rate was a major deterrent to the recruitment of skilled employees in Canada.

The NAFTA agreement had made it easy for qualified Canadian professionals to get a work visa for the U.S. upon receiving a job offer there. The U.S. Immigration Department was also considering raising the number of its work-visa limits for highly skilled workers to approximately 200,000 a year from 115,000. It was estimated that there were nearly 300,000 high-tech job vacancies in the U.S., and with the general unemployment rate in the US near 4 percent (its lowest level in 30 years), Canada was having a difficult time retaining domestically trained employees.

In response to intense lobbying from some of Canada's top business leaders, the Canadian government set up a temporary foreign work program to make entry of skilled workers into the country easier. However, a survey of companies conducted by the Conference Board of Canada discovered that few executives were adequately aware of the program. The federal and provincial governments also started to move toward providing more generous tax treatment of stock options and capital gains. Paul Martin, the Finance Minister, announced the lowering of the inclusion rate for capital gains somewhat modestly, from 75 percent to two-thirds over a period of five years. The federal government also began to allow up to $100,000 in stock options to be exercised every year with tax being paid only when the shares were sold. Previously, tax was triggered when the option was taken up which often resulted in employees having to sell their shares in order to pay the tax imposed on this source of income.

Acquisitions could also create other problems that effected employee's satisfaction and motivation. Most people from acquired companies would experience "title deflation". Typically, a director at PMC would be called a VP elsewhere. Then there was the wage difference between Canada and the US caused by the depreciation of the Canadian dollar in 1998. Salaries at PMC were being raised to try to be more on par but they were not growing as fast as in the US. Most acquisitions were financed with stock options and the people who had stock options in the acquired company got those swapped for PMC stock. As a consequence of the acquisition logic, some employees in a previously acquired firm could have a higher salary and have more stock than the PMC managers to whom they now reported. By 2000 half of the engineers who worked for PMC were millionaires from their PMC stock and so small differences in salary levels had not become too big a motivation problem. However, there was the looming problem of whether people who had so much money would continue to have the same motivation and be willing to work the same long intense hours they did before. There was the additional problem of what might happen if PMC's extreme valuation crumbled.

Organizational Design 1998-2000

PMC entered 1998 with two separate business units. One centered on the company's initial business in ATM, designing and selling WAN related switches. The other one (BIT) had been purchased a year earlier and focused on making chips for the high end Ethernet market (See Case A for a full description). As the effects of the Asian economic flu receded, PMC's core business began to pick up but the company was experiencing internal divisions from the very different styles and operating cultures of its two business units. In addition, there were some internal tensions as some PMC engineers believed that BIT was being given more financial support and more leeway than PMC's traditional business.

Strategically, PMC could see no faster way to grow than to take advantage of its super hot market by continuing to buy small start-up firms. Yet its experience with BIT made it much more reflective about how to do that effectively. One of its next acquisitions was Integrated Telecom Technology ("IGT") in exchange for approximately $55 million (roughly half in cash and the other half in PMC common stock and options to buy PMC common stock). This acquisition expanded their portfolio of ATM layer and switching products. IGT made ATM switching chipsets for WAN applications as well as ATM segmentation and re-assembly and other telecommunication chips. Moreover, IGT was a fabless semiconductor company that contracted out the manufacture of the chips it designed, as did PMC.

Managers in the Burnaby headquarters of PMC agreed that the solution to the biggest problem of ensuring that the integration of acquisitions worked well was to aim for a similar culture. As Kevin Huscroft put it,

> *When we acquire a company that does things differently you get that "our way versus their way" problem. If the engineers are quality people, however, they recognize that in each other and its less of a problem. For the people in the acquired company, they also go from having been at headquarters to now being at the periphery. The way we try to integrate their culture into ours is to get people to move back and forth – but that's difficult to do from a practical standpoint.*

Because IGT had its headquarters in Gaithersburg, Maryland and a development site in San Jose, California PMC faced the question of how to integrate this acquisition into its existing organization. Rather than leave IGT as a separate business, the operations of this acquisition were immediately integrated into the R&D and Marketing groups. Coordination of these two functions for IGT were now performed through the existing organizational arrangements in PMC's headquarters. But geographical distance, different cultures, and different ways of doing things sometimes made coordination and management of disparate product development engineers a problem. That problem was compounded by the increasing need to speed up product development.

Further geographical dispersion of PMC's operations also occurred because the firm was reluctant to lose its experienced system architects for 'personal' reasons. One of PMC's best chip designers, who had come to Vancouver while his wife went to university, wanted to return to his home, Montreal. Not wanting to lose this scarce and most valuable resource to one of their competitors, PMC asked him to set up and staff a design centre in Montreal. This strategy was quite successful as one of his initial designs became an accepted product by the builders of the Internet. Within three years there were about 50 people working in the Montreal office. Managers back in company headquarters in Burnaby found that coordination with the Montreal office was much easier than with other remote locations. Since the manager of the Montreal office had spent two years in Burnaby and knew the "PMC way", many small problems, miscommunications and irritants that occurred with other sites did not happen with the Montreal office.

As the size of the R&D department grew from both acquisitions and a high rate of hire of new employees, a decision was made to separate it into two different divisions. PMC was still concerned about being able to adapt quickly to rapid changes in product markets manifesting new technological developments. In particular, they wished to deploy product development engineers where they were needed. They wanted to avoid building the specialist silos and turf battles between different groups that PMC's founders had experienced in larger corporations. But the assignment and supervision of hundreds of employees was too much to handle by one senior manager. So, the group was split in two.

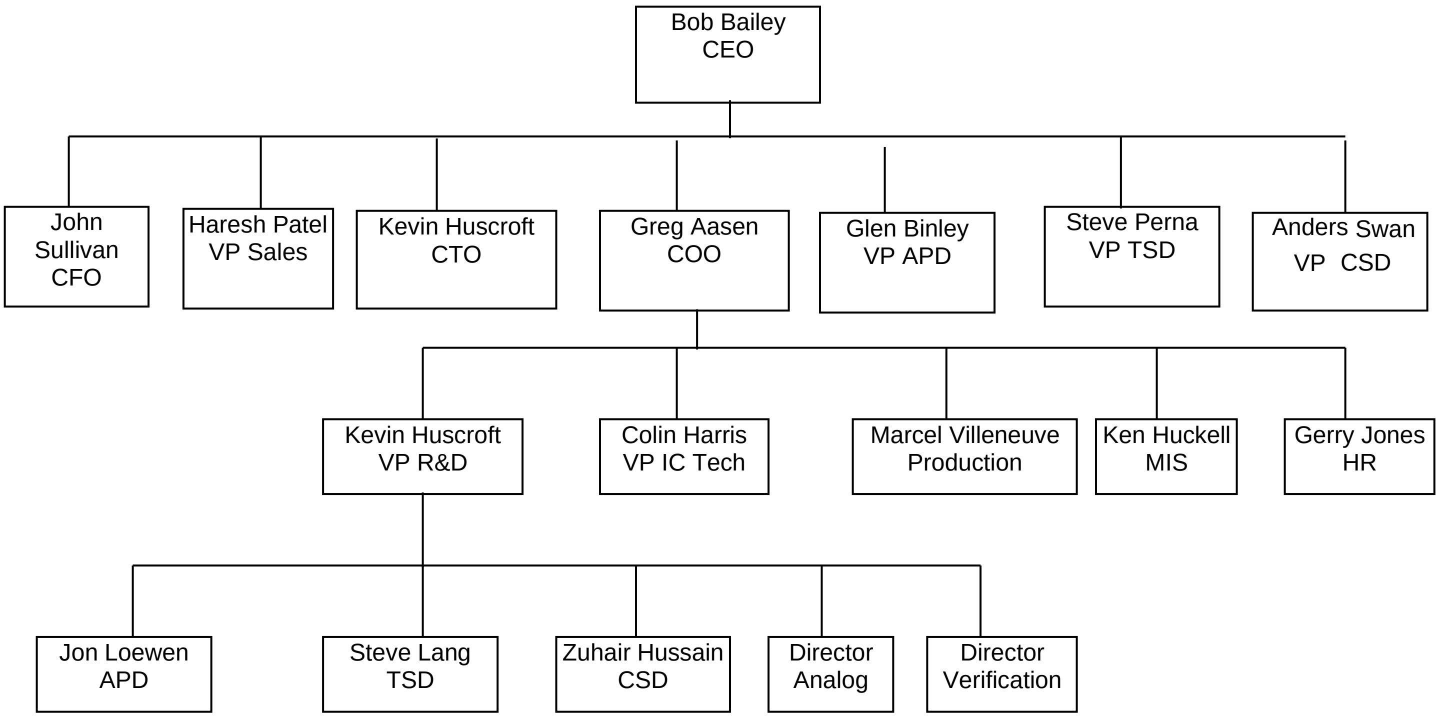
It was decided to organize the critical function of product development by end applications. One division, that came to be known as access products, focused on applications at the very edge of the network where lots of data is coming into one customer and lots of separate streams of data (e.g., lots of different people sending emails) are going out. The other division, transmission switching, dealt with applications that aggregate these separate streams of data up to hundreds of gigabits to be sent down the telecom networks. This proved to be an effective way of unit grouping. Two project managers were informally put in charge of each group with 100 or so engineers dedicated to each group. Marketing, in turn, was informally divided up into those two areas as well with a marketing manager for each area reporting to a common manager. In addition, layout and verification functions were centralized under the IC Technology label.

Within the networking industry, networks were organized in a relatively hierarchical manner in order to scale and manage data flow. At the very edge of the network, a customer might send voice and data to thousands of different destinations. The next layer of interfaces – they could be customers or they could be network to network connections – would handle data going in hundreds of different directions. Following from the logic of this way of grouping functions, PMC's next acquisition, Abrizio, was a company that was developing products at the highest end of this hierarchy, where data was being sent in just a few directions. This provided PMC with expertise in a technical market niche in which PMC had not yet established a presence. Abrizio was acquired in exchange for approximately 4,352,000 shares (approximately US $400 million) in PMC-Sierra stock, warrants, and stock options. Located in Mountain View, California, Abrizio was also a fabless semiconductor company specializing in broadband switch chip fabrics used in core ATM switches, digital cross-connects, and terabit routers. The systems using Abrizio's scalable switch fabric chipset are designed to support POS, ATM, Frame Relay, voice, and IP traffic. The Abrizio TT1 chipset had been well received by the marketplace, having been designed into several Tier 1 broadband customers' mainstream systems.

While PMC determined it made most sense to integrate the design engineers from Abrizio into PMC's R&D group, they did not wish to lose the particular skill set or synergies that had been created and currently existed at Abrizio. Two factors influenced the manner in which Abrizio was integrated with PMC. First, the separation of product development into two groups a year earlier had been relatively successful, partially because PMC's sales and marketing people found they almost never overlapped in their customers or applications. Nevertheless, overall responsibility for all marketing was channeled through one person. Because this was becoming a bottleneck, the marketing function was also formally divided along product lines into the Access Products Division (APD) and Transmission Switching Division (TSD), each headed by a Vice-President. What had previously been an informal division in the R&D group was also formalized with further separation and specialization in the product development area. Most of the engineers from Abrizio were placed in a third R&D group called the Carrier Switching

Division located in Mountain View, California and the former president of Abrizio was made the Vice President of Marketing for Carrier Switching.

PMC-Sierra's Organization January 2000



Although the product development and marketing functions now had parallel structures, the Abrizio acquisition led to further dispersion of the layout function. With prior acquisitions, PMC had either absorbed acquired layout groups into their IC Technology group or had disbanded and laid off the employees previously performing that function. However, in the Abrizio case, the layout group there was retained in their Mountain View location, dedicated mainly to the Carrier Switching Division. Managers were less sure this was a good decision. Layout is an extremely capital and learning intensive business. Each layout designer requires $400,000 in equipment support which created a concern in how effective and efficient PMC was in their deployment and use of those resources. In addition there was the issue of how to enforce corporate standards at remote locations if they were not reporting up through a centralized, functional structure. These standards were evolving every month and were changing to a completely different paradigm every 6 or 7 months.

PMC instituted a cross functional team approach for product development. Once a business plan was accepted at the Project Planning Meeting, a team was created composed of personnel from one of the three R&D units, their associated counterparts in Marketing, and personnel from IC Technology. This was an attempt to speed up time to market. So many issues involve trade-offs between product engineering, product development and marketing. The work flow in the company looked something like this:

Marketing ⟺ R&D ⟺ ICT ⟹ Production ⟹ Sales

though in reality much of the work flowed in parallel processes with members of marketing, R&D and IC Technology being involved at all stages from concept to pre-production. Every three months, at the Product Management Team Meeting, they decided how best to reallocate these resources. They found that, after the change, the time from sign-off of specifications and parts was still the same but the teams allowed them to improve on the back end once the prototype was developed. PMC prided itself in being quick on time to market and as a result design engineers would work long hours and weekends as deadlines approached to meet their due dates. During 1999 as PMC's stock rose exponentially, many of these employees seemed happy to be contributing to the increase in value of their own vested shares. As CEO Bob Bailey put it, "It's like there is money lying all over the ground and we are running as fast as we can to pick up as much as we can".

As PMC continued to absorb new acquisitions, one prior acquisition, BIT in Portland, proved to be no longer supportable. Problems with BIT continued and this acquisition was unable to be successful in the ethernet business. In October 1999, PMC closed down the BIT business unit and approximately one third of their Portland employees were laid off. The work teams associated with those employees still remaining in the BIT organization were broken up and re-assigned and re-integrated into PMC's R&D group in an attempt to overcome the hard feelings on both sides resulting from the lack of success. PMC tried to capitalize on the ethernet expertise of their remaining BIT employees and assigned them to products for WAN systems such as Cisco routers. By 2000, many of the engineers in the Portland office were relatively new to PMC - but there was not the same level of productivity and morale there as there was in the company headquarters in Burnaby.

Despite PMC's mixed experience with its ability to integrate acquisitions into its existing operations, the strategic logic associated with its huge market evaluation continued to indicate future acquisitions. A decision was made to acquire companies that filled in the gaps in PMC's product line as the company moved from the design of single chips to the design of suites of chips.

By the Spring of 2000 PMC had operations in Saskatchewan, Quebec, Ireland, Pennsylvania, Oregon, Maryland, two in Ontario and four in California. PMC had grown from about 500 employees at the beginning of 1999 to about 1,000 a year later and all indications suggested that PMC's workforce would continue to grow at that rate. Over 100 job openings were posted on its website. The R&D (chip design - product development) group alone had close to 500 employees, with over 200 employed in TSD, close to 200 employed in APD, about 50 in CSD and the rest employed in verification and analog. R&D had huge scope as employees there were working on close to 100 projects. In the past, Product Planning Meetings were less than a dozen people. Now however there could be more than thirty persons in the room. Not surprisingly, PMC managers wrestled with problems of how to best organize the company, absorb growth, integrate remote locations and maintain motivation and productivity while competing in a market that was rapidly changing. Steve Lang:

> *We need managers who have strong opinions about, in a given area, what the products that are good to pursue and what the products that are not good to pursue and be able to debate actively the pros and cons of putting out limited resources to work on something versus another — and I can't be expected to do that for every project in a 300 person organization. I used to be able to do it for every project but I can't now. We're getting too big for our management system.*

Geographic dispersion contributed to at least two organizational problems. One concerned the difficulty of obtaining corporate-wide coordination over current and emerging projects. In particular, the forward-looking activities of Strategic Visioning Teams that 'sensed' new market opportunities required moderately frequent face-to-face meetings. Getting product development choices correct was a critical component of PMC's future performance. If it assigned significant resources to the development of products that generated few or no sales – then the organization would be compromised. Despite efforts to make use of video conferencing, PMC's managers found that the added costs didn't bring much in benefits over simple voice conferencing. In addition, PMC managers had tried many of the internet based "collaboration tools" available and had found them lacking. As TSD R&D Director Steve Lang put it, "the best collaboration tool is an airplane ticket".

For more structured tasks such as the assignment of different components of a multi-layered chip to different design groups, coordination was less of a problem. The standardization associated with the Telecom industry allowed the delegation of responsibility and accountability for the design of fairly independent parts of a chip to different groups that could work relatively autonomously from each other. As Steve Lang described it,

The best way of managing coordination is to partition the functions in a way that we minimize the communication that's required between the design centers. So what I mean by that is in a chip, a big complex thing, you would partition areas, spec out blocks. Once those block specs are solid with respect to what interfaces are, what functions it has to do, then that stuff can not only be farmed out to the different design centers, but also to sub-contractors. And that's worked reasonably well. So when people talk about doing it by different locations, they're not talking tasks that require a lot of communication.

There can be issues that arise when we have to put those blocks together. If it happens its usually an interaction of features, interaction of behaviours that causes a problem and that could be traced back to an error in the way the block was initially specified or in its implementation and would need to be addressed via meetings maybe, plane flights, telephone calls, whatever. It falls back to that. What I'm describing here is not a panacea. Ideally the collaboration tools would be better and you'd be able to interact much more and not have to care so much or pay so much attention at the front end.

By the way, this partitioning I'm describing is not unique to the fact that we're geographically dispersed, we've always done this from day one even when we were all located here in Burnaby. To divide and conquer and break things up so that blocks can be given up to designers. It's just that when there are geographic boundaries other more cultural issues can also come up and it becomes a bit more difficult to deal with.

One aspect of the PMC-Sierra culture that seemed to stand out in the minds of Burnaby managers was the strong technical background of marketing personnel. Conflict between Marketing, which wants to take its cues from customers, and Research & Development, which is more oriented to what is technically appropriate, is endemic to product development organizations. Conflicts like these were visible in PMC but were generally experienced as healthy. According to Glen Binley, *I would say one of the key reasons we have been successful is that we have always had a very positive and non-confrontational relationship between the strategic marketing people and the engineering people and perhaps that is generally an uncommon thing. I think the reason that we have been able to make that work is that most of the marketing people in this organization came out of product development. They have strong technical background and so they've been there before and they know what it's like trying to develop things and they don't ask for totally unrealistic things. So they have the respect generally of the people on the development side. I think that's one of the reasons that we have been successful. That we've been both a marketing and technically driven organization. Not just one or the other.*

A similar comment was made by Steve Lang,

Inside PMC in Burnaby, the different managers or directors of these design groups have come up largely through the technical ranks and the model we have now is that anyone who is a manager or a director is very strong technically. He's done multiple devices in his or her career and is very strong on the product side and still has huge influence and impact on what we do and how we do it. Whereas often the other companies that you acquire aren't built that way and the people you would see in the positions of directing or running these fifty person groups are much more classical managers, perhaps, who are more involved with the administration and interested in career planning. So what you see is that the core values of the people and how they've gotten to the point that they gotten to, say in Burnaby, are quite different and it's really hard to make the two work together is what I've found.

The issue of overall technical competence was an imperative for both strategic choice and the integration and coordination of projects. Few, if any, non-technical managers could fully appreciate the opportunities, synergies and interdependencies among projects. Technical competence, specialized knowledge, and a common culture were regarded as essential features of a successful organization in the networking industry. As Kevin Huscroft commented:

You want to have some product focus in R&D groups – you want them to feel more identified with the product. On the other hand, R&D people can have some issues working for marketing types, especially if the business manager isn't technically strong.

They're afraid the marketing types may commit your company to do something that isn't technically feasible. If we acquire companies that have close R&D and marketing we don't want to rip that up. As well, my directors are pushing back and saying they don't want another remote site reporting in to them. I think it's better to have a group who feel ownership for dominating a business/technological space. One view is that we keep the back end (IC Technology) strongly centralized to maintain standards and let the rest work out however it does.

By 2000 a logic of acquisition was emerging among managers in PMC. Some acquisitions were made to increase the design capacity of PMC while others were made to fill holes in product portfolios. PMC managers had noticed that it was easier to manage the latter because they could be left to operate more independently. The former founders running those organizations tended to have strong technical skills and were able to interact with their marketing counterparts in a way that ensured the right tradeoffs for the products. They could be brought into the organization as a remote unit that, for the most part, ran autonomously. This was also effecting how managers saw remote units that did not have a separate area of expertise. Managers believed that remote units were happier and more productive if they had a product, and area of technical expertise, that was their own and they were looking for ways to create that in all remote units.

But the advantages of being able to slice up the product portfolio of PMC into smaller, self contained units faced some challenges, not the least of which was the uncertainties they faced in their technology and markets. Silicon did not seem to be able to keep up with bandwidth demands that were doubling every six months. The convergence of chip technology was leading to a possible need to converge skill sets that currently were in different divisions.

Glen Binley:

I definitely have a goal of trying to organize things in that way and so that you can continue to silo things and give people, "ok you focus on this and make this happen." But at the same time people have to be adaptable to the reality that silos may collide and need to be integrated. When we buy something, if they have a function, a product focus that makes logical sense to stand alone and evolve in its own right, then that makes perfect sense. The problem you get into is where, from a competitive perspective, it makes sense to integrate two pieces. We can't organize ourselves to be convenient for us, we have to organize ourselves to meet the needs of the marketplace and there are *certain products or technologies that will inherently get integrated over time. Those are the ones where you have to be very careful that organizational structures don't stop you from doing the right thing from a competitive and a customer perspective.*

By the summer of 2000 it looked like the strategy of integrating new acquisitions into the functional/business unit structure of PMC needed to be retooled. But this was a difficult issue to manage. As VP of IC Technology Colin Harris put it,

All aspects of organization pre-occupy us – almost all of the time. If we don't get it right, careers can go up in smoke. We have trouble talking about possible alternatives to our current form of organization. You just can't, in a public meeting, go to the chalk board and suggest a new form of organization which will change radically the work of someone who's in the room. This is very, very difficult. These difficulties will increase as we go. The boundaries were much less visible when we were younger and smaller.

Greg Aasen, the COO:

Now that our R&D resources are not so thin time to market is our key concern. Creating directors in business units that are driving the product has helped but we still need functional experts to drive the skill sets. One model we are looking at is to break the R&D group and reassigning engineers to one of three business units. This will free up more of Kevin's time to do the CTO function and perhaps create tighter integration of marketing and R&D and thus speed up time to market. There are some downsides to this plan however. Right now the heads of business units have small staffs and can spend a lot of time with customers. Growing the units will drastically increase the administrative load. Another is that by sticking all the R&D resources in business units it makes it more difficult to move into new types of businesses. And there is the further complication of what to do with companies that we buy.

Nevertheless, issues of organizational design and the overall management of the company continued to evolve and pre-occupy PMC's managers as they contemplated their uncertain future.